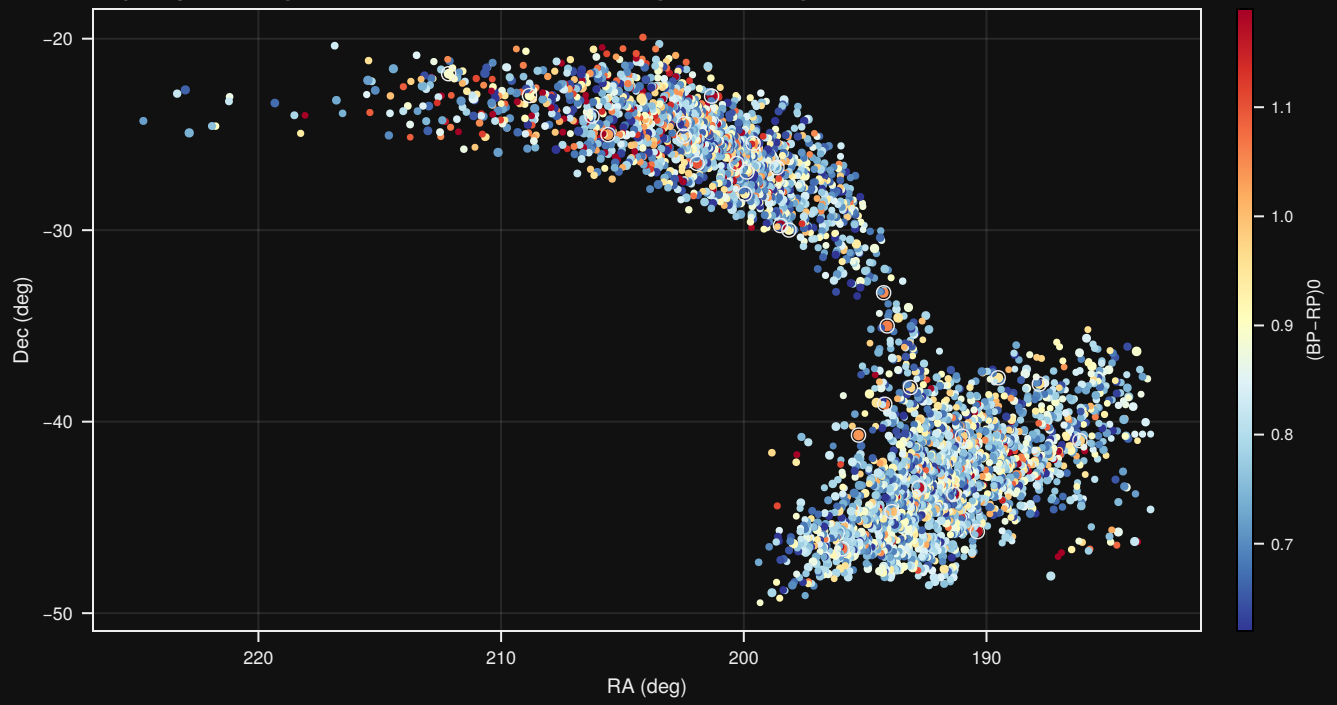
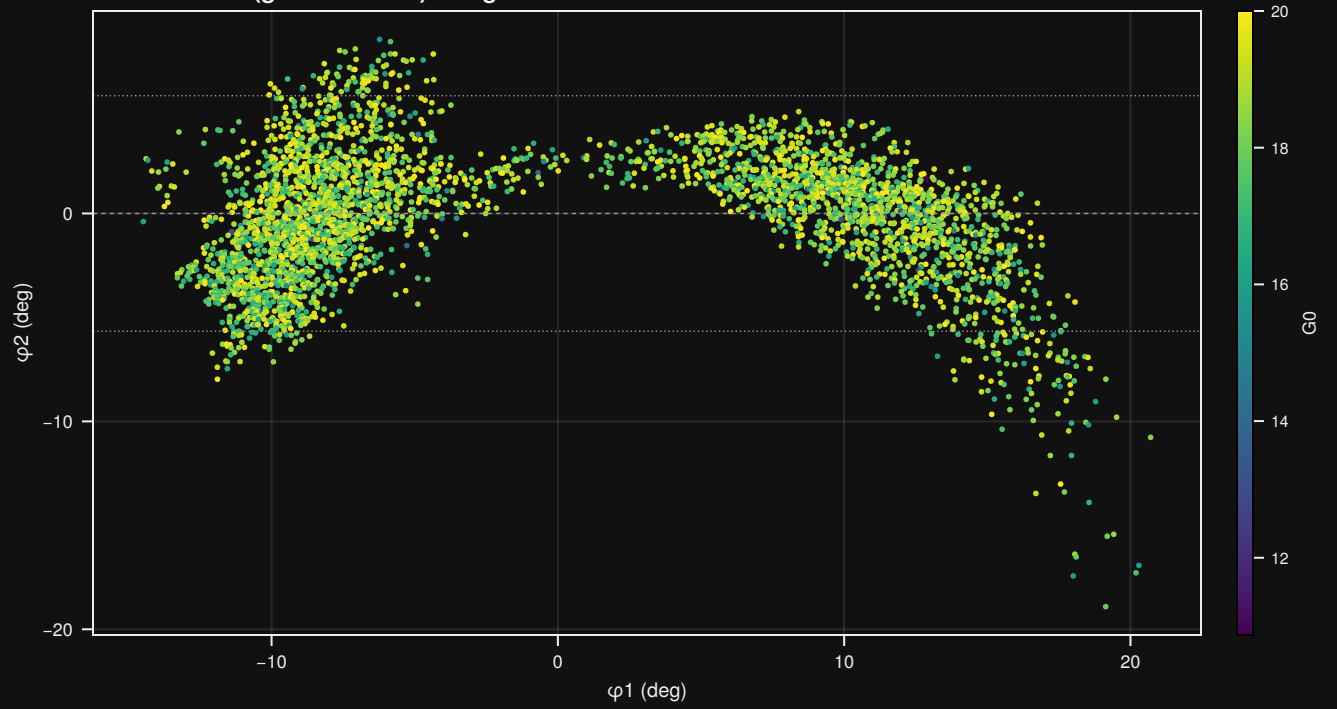
Svol

MEMBERSHIP	
Stars in catalog	3724
with Gaia GSP-Phot	1990 (53%)
with radial velocity	29
Stream id (sID)	54
GEOMETRY · great-circle fit	
Centroid RA, Dec	196.2, -38.6 deg
Centroid l, b	307.0, 24.1 deg
Pole RA, Dec	123.1, 21.8 deg
Length (φ1, 2–98%)	28.6 deg
Width σ(φ2), robust	2.83 deg
KINEMATICS	
pmRA* med ± σ	-5.52 ± 2.95 mas/yr
pmDE med ± σ	-7.15 ± 1.70 mas/yr
μφ1* med ± σ	-8.79 ± 2.54 mas/yr
μφ2 med ± σ	1.30 ± 2.07 mas/yr
Parallax (cat.) med	0.237 ± 0.221 mas
Parallax (wtd, ZP-corr.)	0.264 ± 0.010 mas
Distance	d = 3.8 ± 0.1 kpc
Physical length	1.89 kpc
RV median ± σ(MAD)	225.1 ± 24.7 km/s
RV gradient dV/dφ1	-1.66 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	10.9–20.0 (18.7)
(BP−RP)0 median	0.80
GSP-Phot [M/H] (n=1990)	-1.66 ± 0.56
GSP-Phot Teff median	5550 K
RUWE > 1.4	1.1 %
PM clusters (largest)	260 (42 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	3724 | 234
n_RV catalog | paper	29 | 16
<[Fe/H]> paper	-1.98
Stellar mass (paper)	2000 Msun
Pole offset vs paper	73.0 deg
LABEL CHECK	n differs from paper!

Sky map · catalog members (n = 3724) · size ∝ brightness · ring = has RV

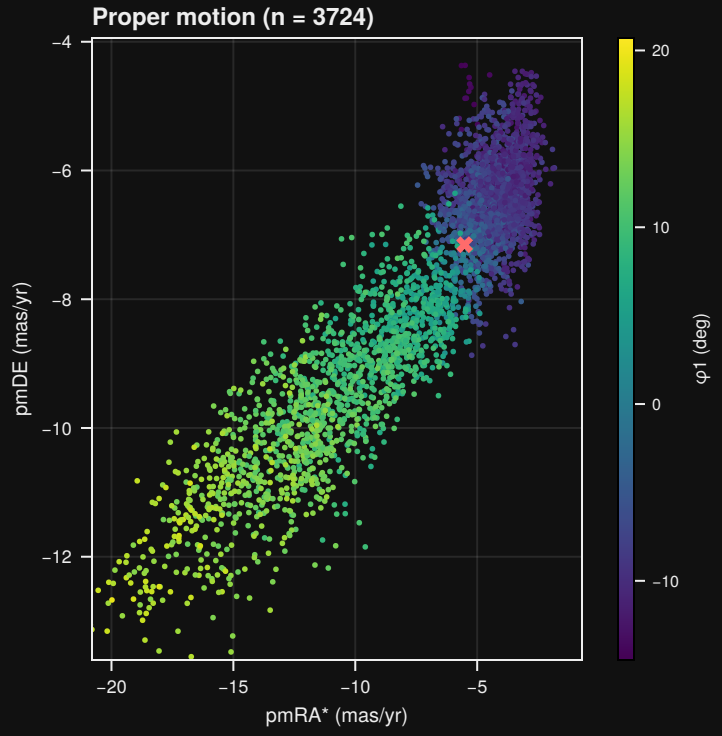


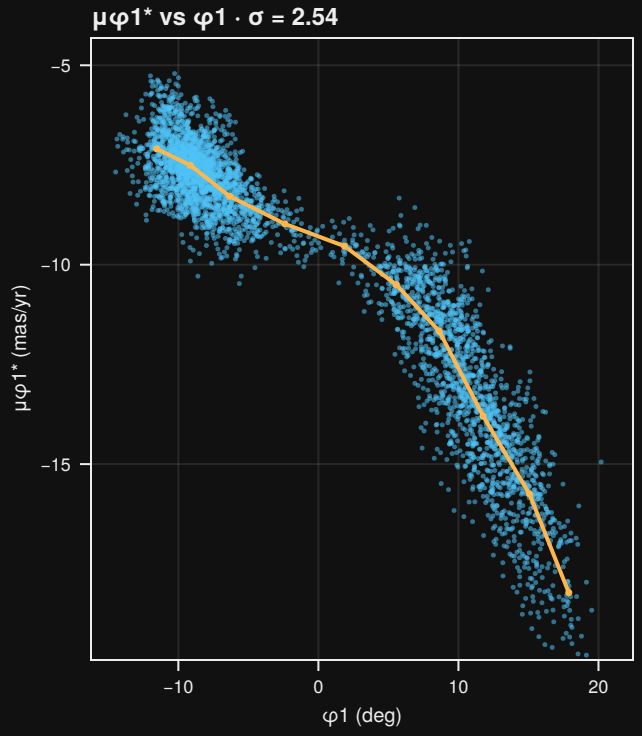
Stream frame (great-circle fit) · length 28.6° · width σ 2.83°

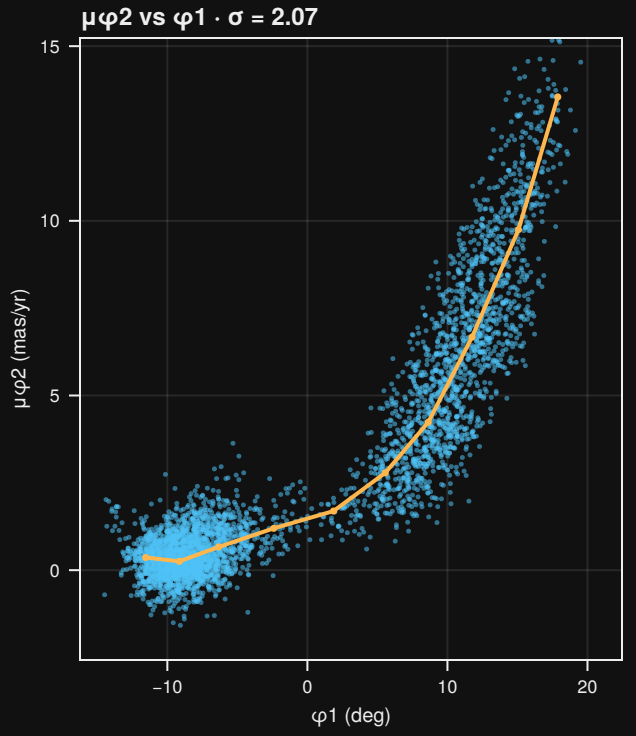


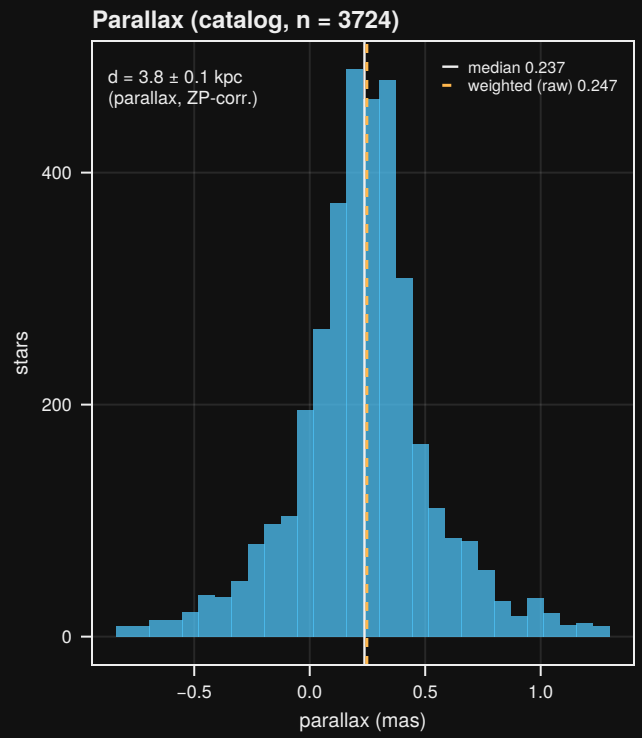
Svol

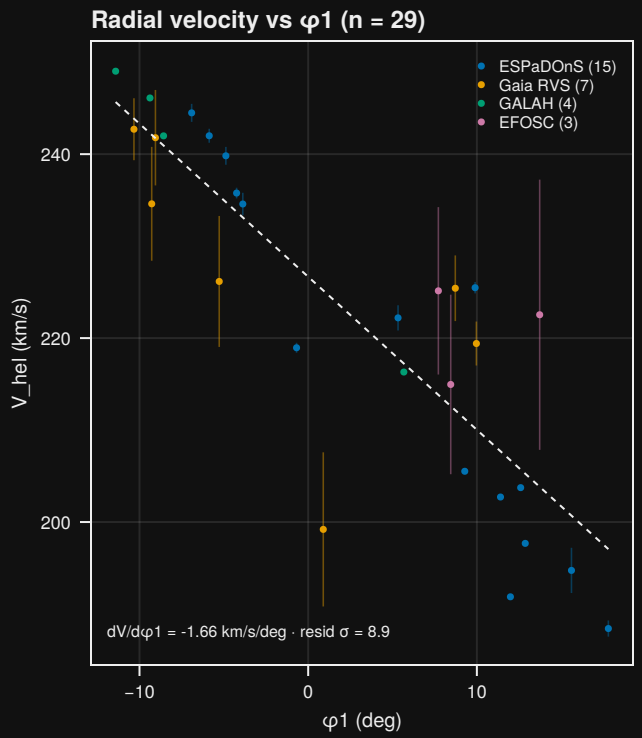
Kinematics · proper motion · parallax · radial velocity

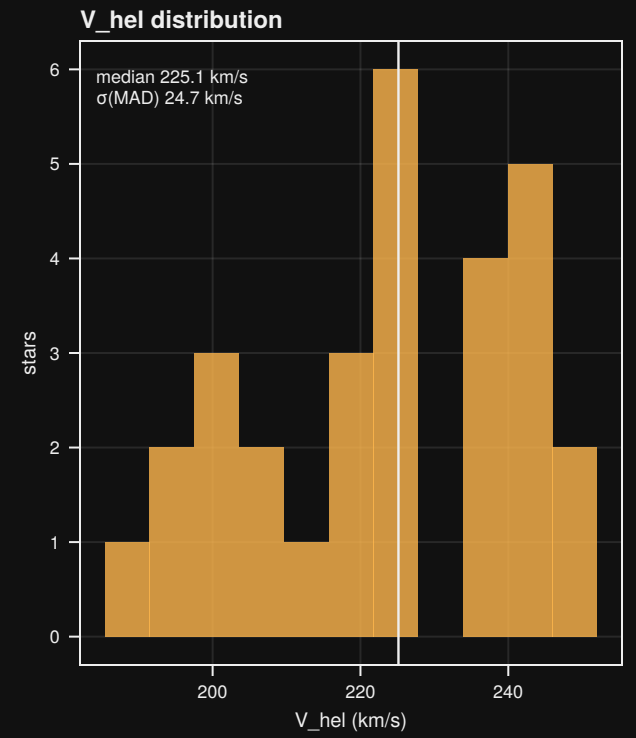






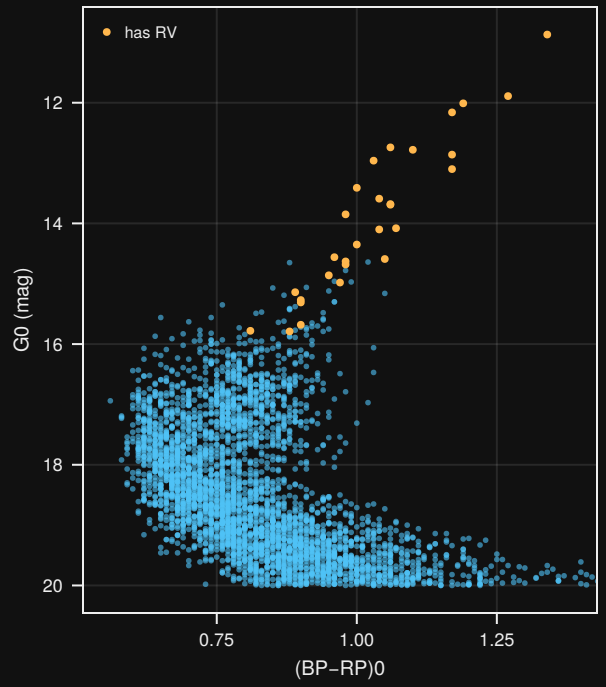




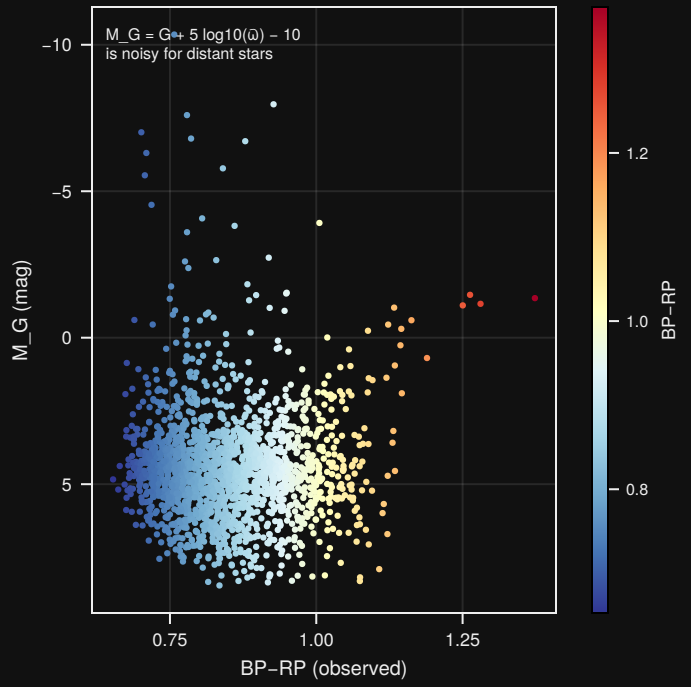


Svol

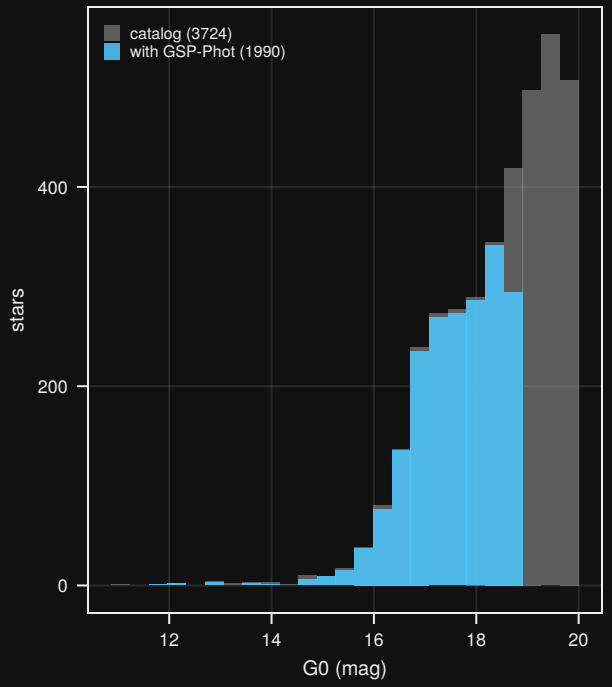
CMD · dereddened (n = 3724)



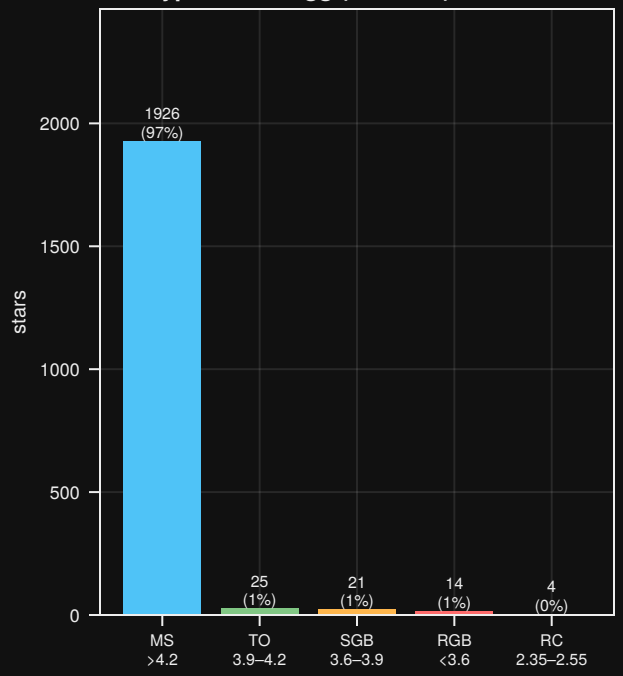
HR diagram · ϖ>0 (n = 1838/1990)



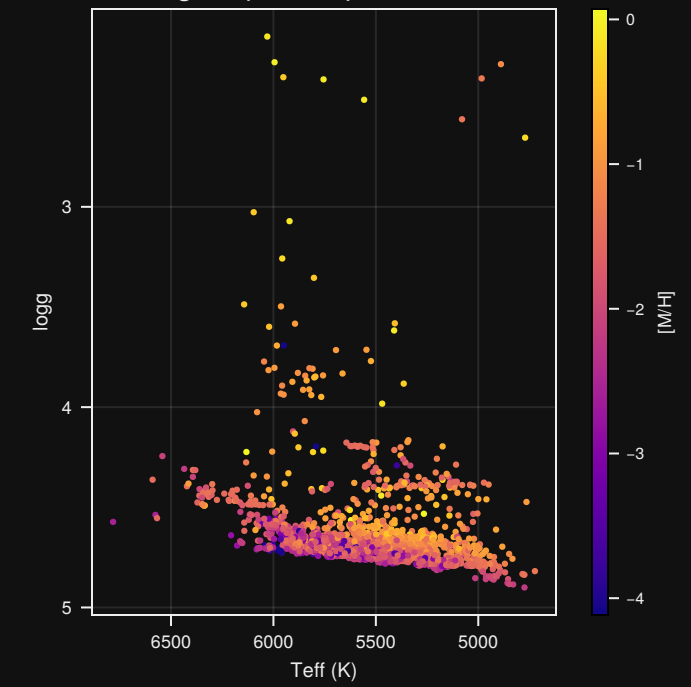
G0 & GSP-Phot coverage



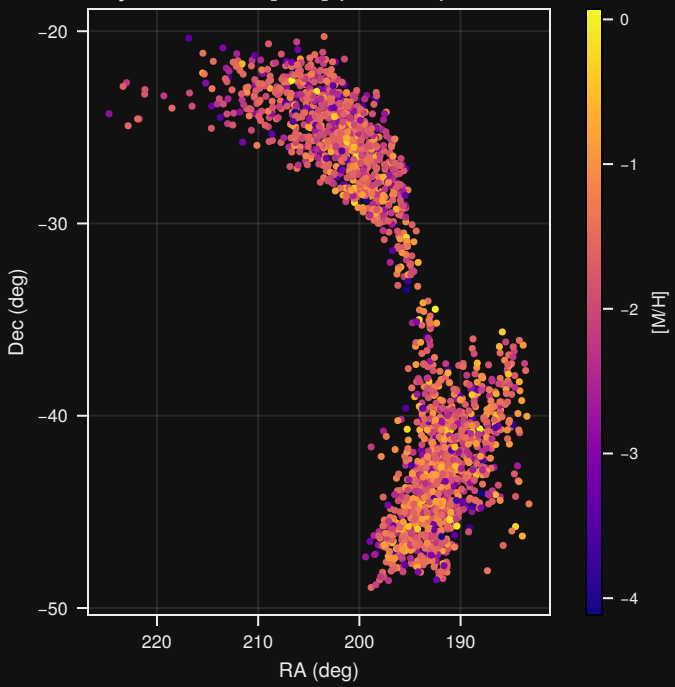
Star types from logg (n = 1990)



Kiel diagram (n = 1990)



Sky · GSP-Phot [M/H] (n = 1990)

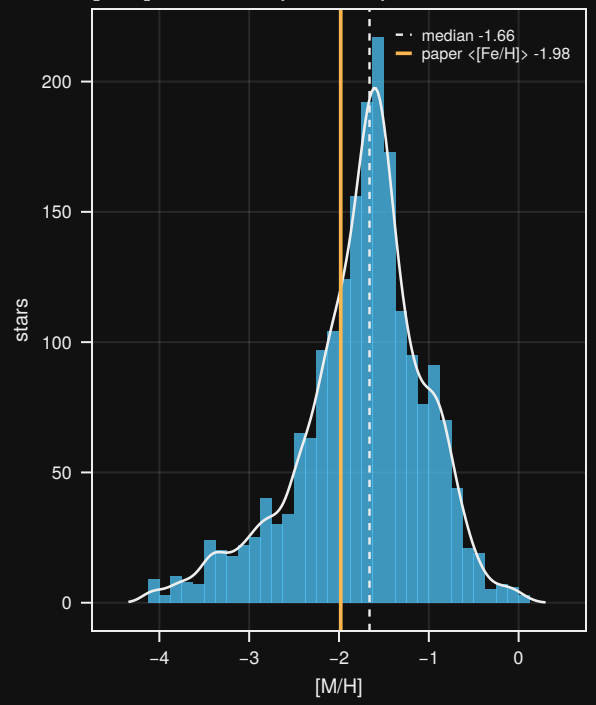


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

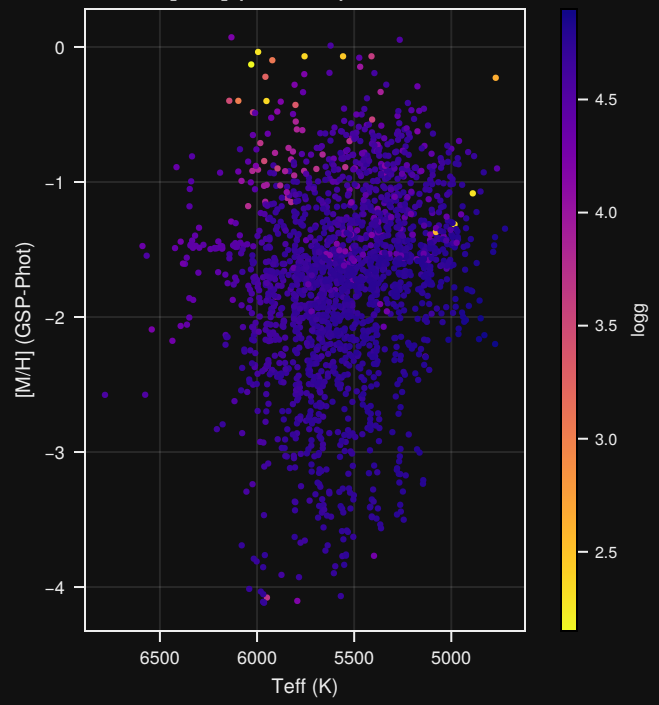
Svol

Metallicity · age · astrometric quality

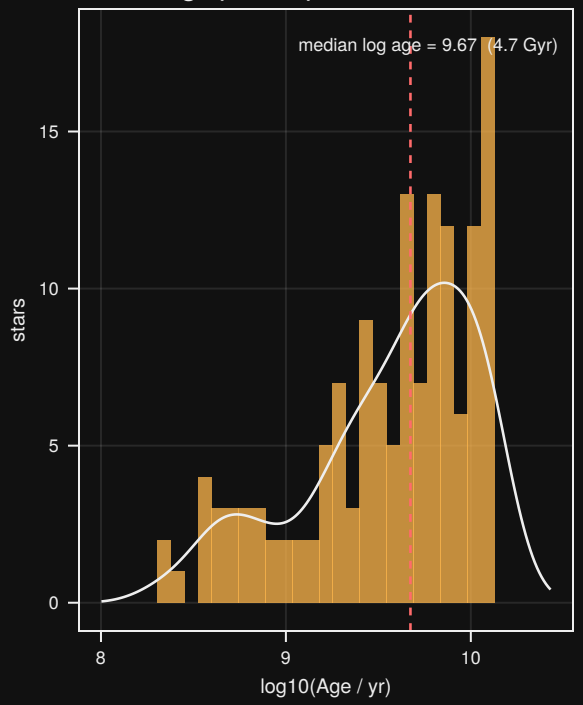
[M/H] GSP-Phot (n = 1990)



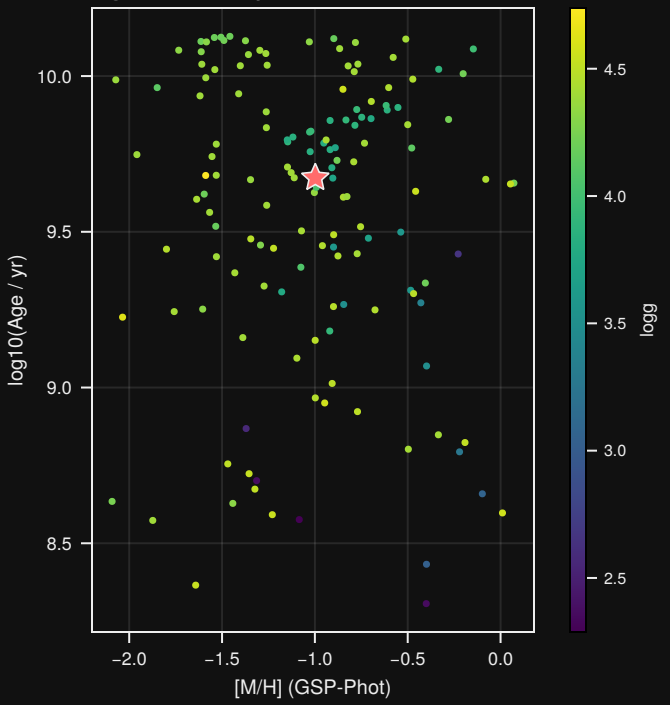
Teff vs [M/H] (n = 1990)



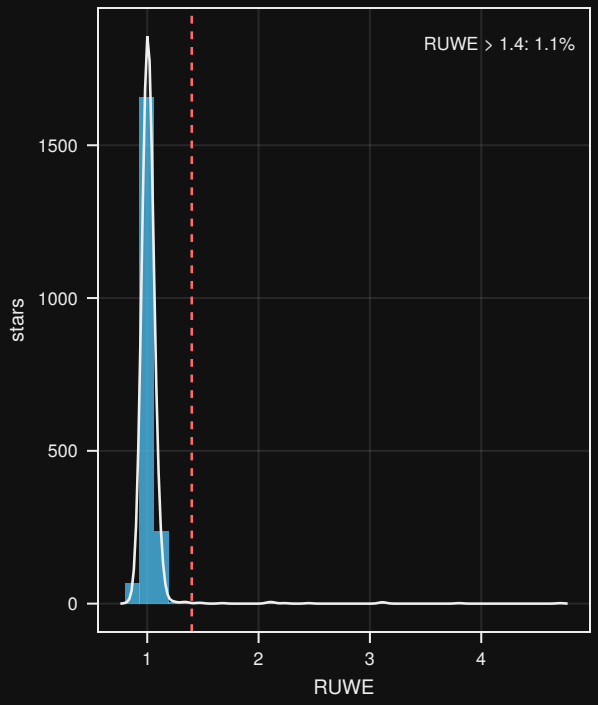
FLAME age (n = 145)



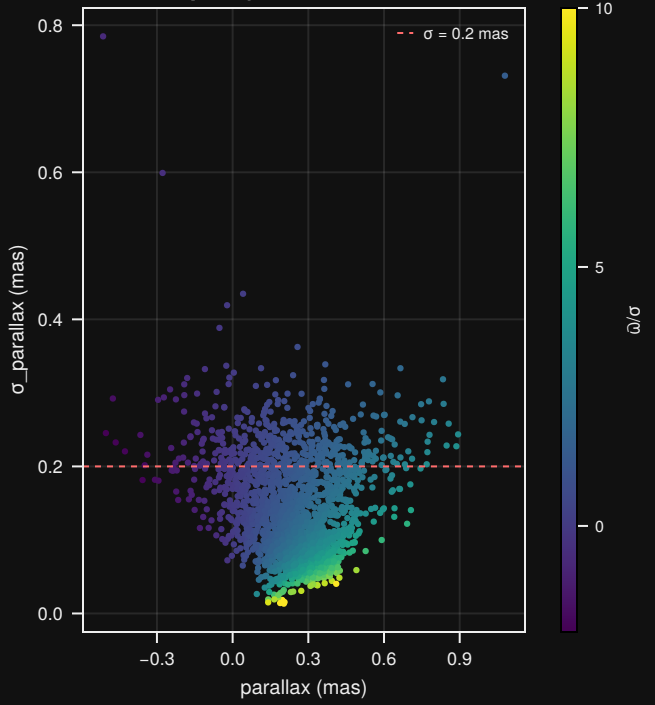
Age–metallicity (n = 145)



RUWE (n = 1982)

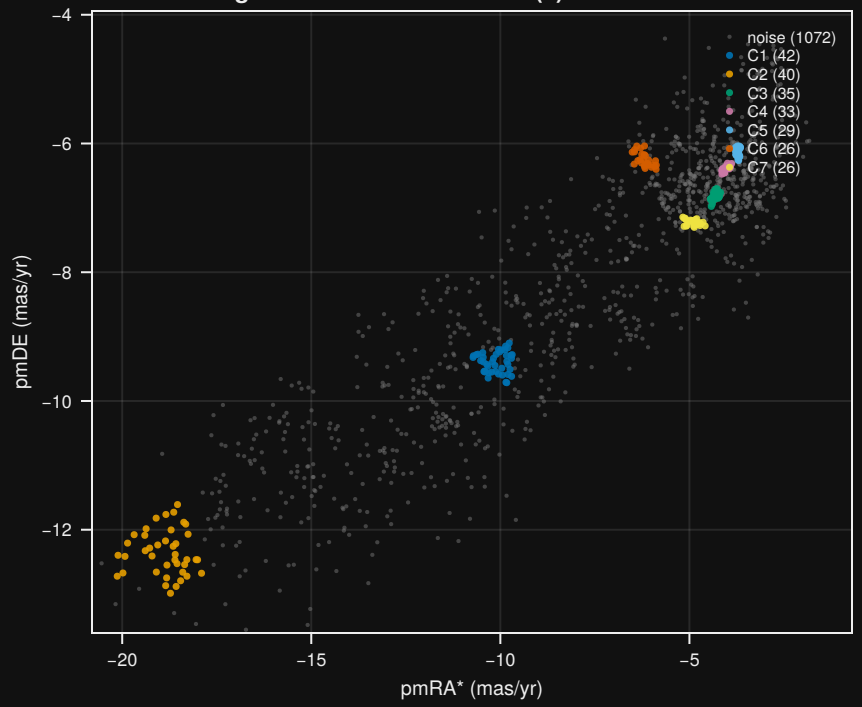


Parallax quality (n = 1990)

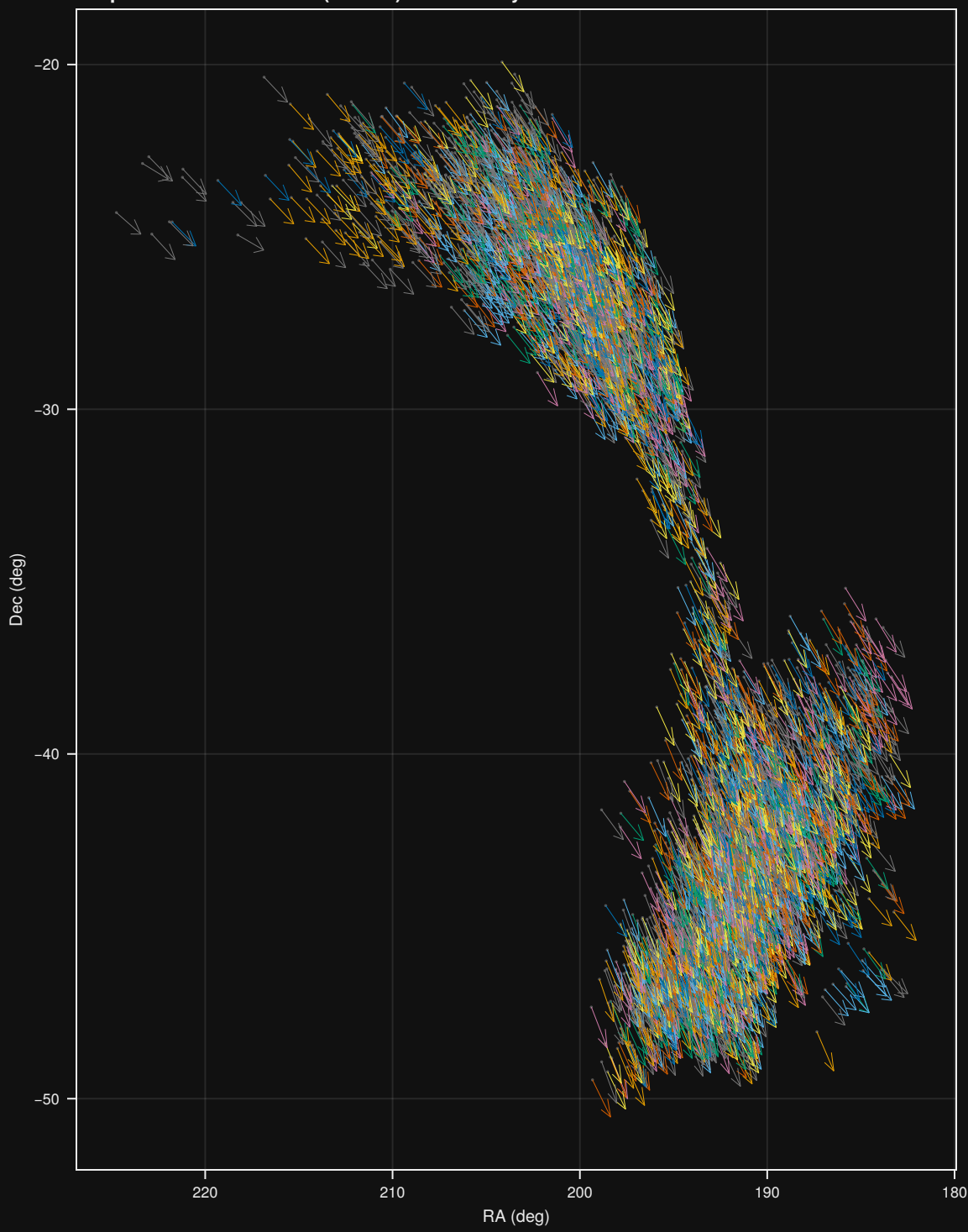


Svol

PM clustering · HDBSCAN · 260 cluster(s) · noise 29%



Proper-motion directions (arrows) coloured by cluster



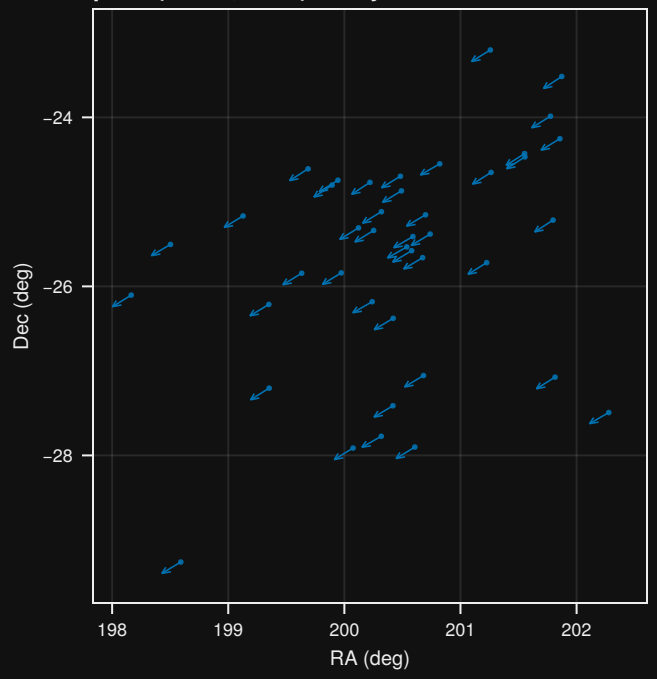
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	1072	-6.00	-7.38	3.85	-1.67	9
C1	42	-10.08	-9.39	0.39	-1.82	0
C2	40	-18.72	-12.40	0.55	-1.79	0
C3	35	-4.31	-6.82	0.09	-1.93	0
C4	33	-4.06	-6.40	0.07	-2.10	1
C5	29	-3.73	-6.13	0.05	-1.64	0
C6	26	-6.19	-6.27	0.17	-1.94	0
C7	26	-4.88	-7.23	0.18	-1.32	0
C8	26	-6.19	-8.47	0.19	-1.86	0

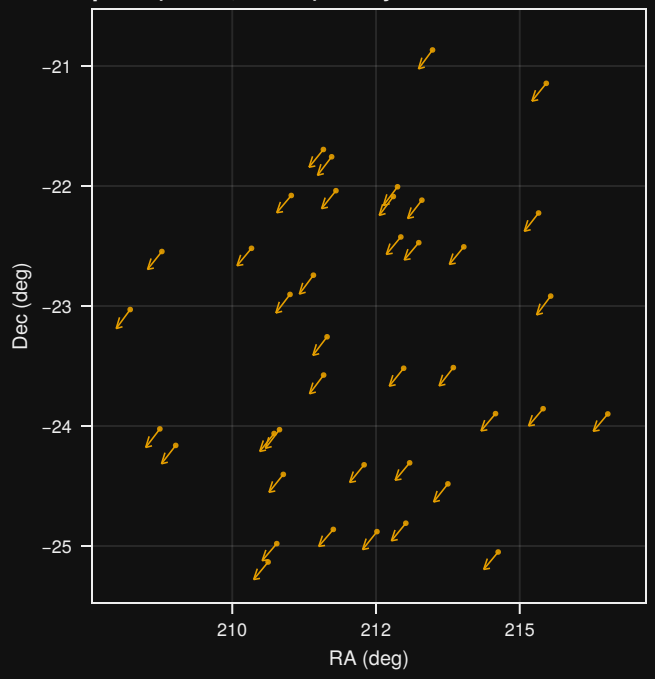
Svol

Per-cluster proper-motion vector maps

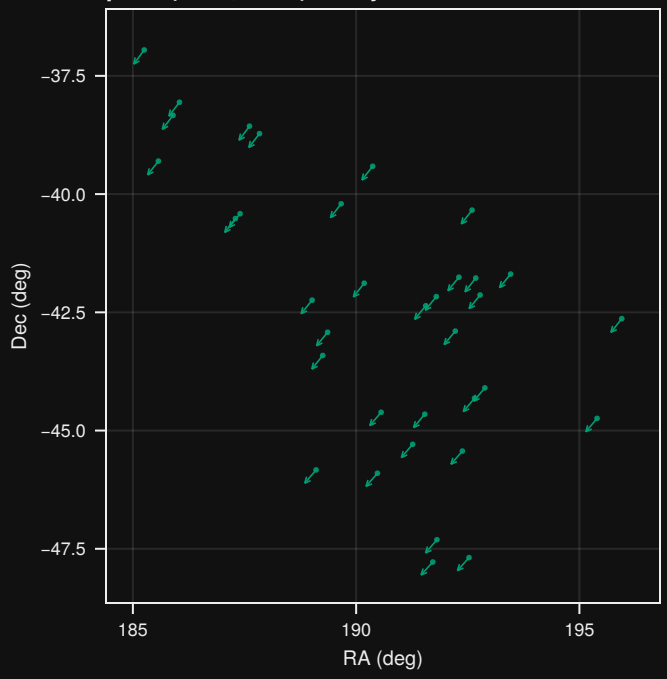
Cluster C1 · n = 42
pm = (-10.08, -9.39) mas/yr



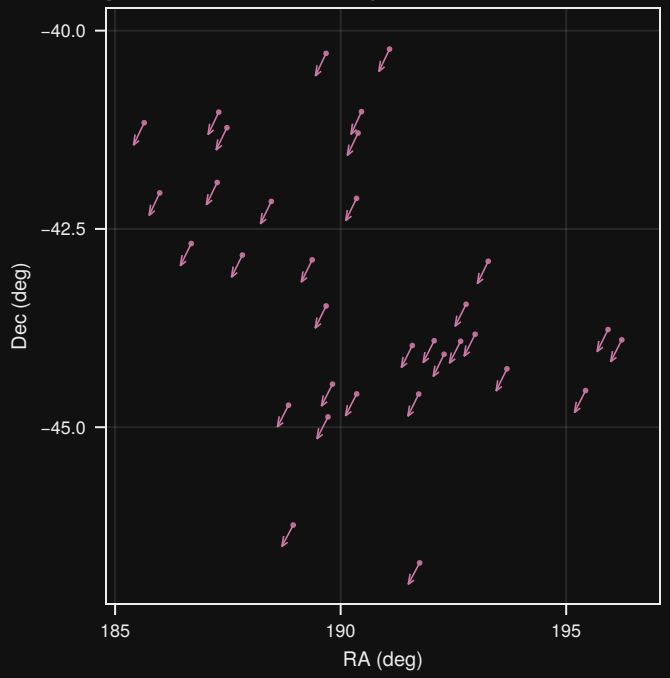
Cluster C2 · n = 40
pm = (-18.72, -12.40) mas/yr



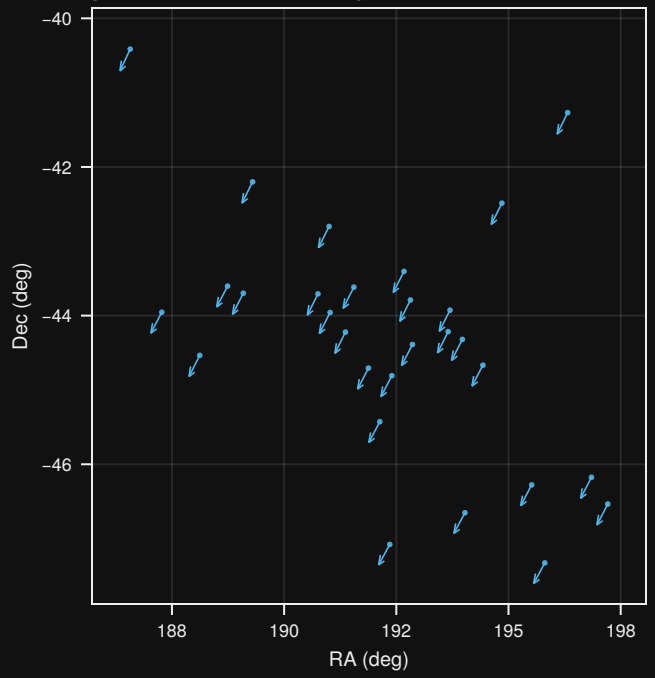
Cluster C3 · n = 35
pm = (-4.31, -6.82) mas/yr



Cluster C4 · n = 33
pm = (-4.06, -6.40) mas/yr



Cluster C5 · n = 29
pm = (-3.73, -6.13) mas/yr



Noise · n = 1072

